moment of the universe, all the spatial dimensions of string theory are on completely equal footing—they are completely symmetric—all curled up into a multidimensional, Planck-sized nugget. Then, according to Brandenberger and Vafa, the universe goes through its first stage of symmetry reduction when, at about the Planck time, three of the spatial dimensions are singled out for expansion, while all others retain their initial Planck-scale size. These three space dimensions are then identified with those in the inflationary cosmological scenario, the post-Planck-time evolution summarized in Figure 14.1 takes over, and these three dimensions expand to their currently observed form.

Why Three?

An immediate question is, What drives the symmetry reduction that singles out precisely three spatial dimensions for expansion? That is, beyond the experimental fact that only three of the space dimensions have expanded to observably large size, does string theory provide a fundamental reason for why some other number (four, five, six, and so on) or, even more symmetrically, all of the space dimensions don't expand as well? Brandenberger and Vafa came up with a possible explanation. Remember that the small-radius/large-radius duality of string theory rests upon the fact that when a dimension is curled up like a circle, a string can wrap around it. Brandenberger and Vafa realized that, like rubber bands wrapped around a bicycle tire inner tube, such wrapped strings tend to constrict the dimensions they encircle, keeping them from expanding. At first sight, this would seem to mean that each of the dimensions will be constricted, since the strings can and do wrap them all. The loophole is that if a wrapped string and its antistring partner (roughly, a string that wraps the dimension in the opposite direction) should come into contact, they will swiftly annihilate one other, producing an unwrapped string. If these processes happen with sufficient rapidity and efficiency, enough of the rubber band-like constriction will be eliminated, allowing the dimensions to expand. Brandenberger and Vafa suggested that this reduction in the choking effect of wrapped strings will happen in only three of the spatial dimensions. Here's why.

Imagine two point particles rolling along a one-dimensional line such as the spatial extent of Lineland. Unless they happen to have identical velocities, sooner or later one will overtake the other, and